

D458

Introduction to Systems Thinking for Health Professionals

Armored Drill

8 concepts · 12 items · 4 transfer items

What this is. Every concept in this course's public description, re-asked with more disguise than a practice test uses. Each stem carries facts that do not decide it. Every wrong answer is a named trap.

Source. Scope from the course's public description in the WGU Institutional Catalog (September 2026); items written from the field's standard body of knowledge. No WGU assessment item, course material or proprietary content is used or reproduced. Not affiliated with or endorsed by Western Governors University.

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How to run it

Paper first. Pencil. One pass, no notes, a timer if you want the pressure.

1. **Before you read the options**, fill the strip under the item: the target word (what is actually being asked), and the one fact in the stem that decides it.
2. **Principle before arithmetic**. Name the rule, then compute.
3. **Camouflage**. Most of every stem does not matter. Ask: which single fact decides this?
4. **Inverse pairs are the bait**. If two options are opposites, one of them is the trap built for you.
5. **Mark sure, unsure or guess** honestly. Sure-and-wrong is the most valuable miss on the page — it is a wrong mental model.
6. **Absolutes** (always, never, guarantee) are usually wrong. And recognizing an answer is not the same as owning it.
7. Items with a **heavy black bar** on the left are **transfer items**: the same principle, reversed or solved for a different variable.
8. Stop at the STOP page. Then grade, fill the triage grid, and tally the traps you fell for. The biggest tally is what you drill next.

The strip, filled in, looks like this:

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	_____	

Items

1.

A rural clinic with a new parking lot replaced its entire front-desk team after long wait times, but waits were back within three months. Scheduling still books every patient in 15-minute slots, and pay bonuses still reward visit volume. The clinic's logo was also redesigned. What best explains why the problem returned?

- A. The scheduling rules and volume incentives, which drive behavior, were unchanged.
- B. The new front-desk staff were less skilled than the old team.
- C. Replacing staff always fixes a problem if given more than three months.
- D. The logo change confused patients about the clinic's identity.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

2.

At a hospital that recently repainted its lobby, nurse burnout leads to resignations, which raise the workload on remaining nurses, which increases burnout further. The chief nursing officer is new to the region. What kind of loop is this?

- A. A linear cause-and-effect chain with no feedback.
- B. A delay, because resignations take time.
- C. A balancing loop, because it produces a bad outcome.
- D. A reinforcing loop.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

3. [TRANSFER — reversed stem / different variable]

Which example describes a balancing loop?

- A. More infected people lead to more infections, which lead to more infected people.
- B. Positive reviews bring more patients, whose reviews bring still more patients.
- C. When a patient's blood sugar rises, insulin release increases and brings it back down.
- D. Any loop that includes a delay.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

4.

A hospital with a highly rated cardiac unit starts Monday with 180 inpatients. For the next 4 days it averages 45 admissions and 50 discharges per day. The emergency department also saw 300 visits a day. How many inpatients will there be at the end of day 4?

- A. 200
- B. 160
- C. 175
- D. Cannot be known without the emergency department visits.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

5. [TRANSFER — reversed stem / different variable]

A unit has 180 patients and discharges 50 per day. What average daily admission rate would raise the census to 200 in 4 days?

- A. 5 per day
- B. 55 per day
- C. 45 per day
- D. 70 per day

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

6.

A health system known for its research grants sees staffing shortages and posts many jobs. New hires take about 10 weeks to start. Seeing no change after 4 weeks, the manager posts even more jobs; three months later the units are overstaffed and hiring is frozen, and the cycle repeats. The HR office moved to a new building last year. What best explains the swings?

- A. The manager's poor judgment; a better manager would never overshoot.
- B. A reinforcing loop that makes staffing grow without limit.
- C. A delay between posting jobs and new hires arriving, which leads to overcorrection.
- D. Confusion caused by the HR office's move.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

7.

A county health department with a popular newsletter wants to reduce missed prenatal visits. Its partner clinics are paid per visit completed. Which intervention is likely to have the greatest leverage?

- A. Change the payment rule so clinics are rewarded for healthy births and continuous care rather than visit counts.
- B. Hire more clinic staff, since more resources always outperform rule changes.
- C. Add a missed-visit reminder to the popular newsletter.
- D. Increase the reminder-call budget by 10%.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

8. [TRANSFER — reversed stem / different variable]

Which intervention is the lowest-leverage type of change?

- A. Shifting the organization’s goal from treating illness to keeping people well.
- B. Raising the number of flu-shot clinic hours from 20 to 24 a week.
- C. Changing the rules for who can order tests.
- D. Making infection rates visible to every unit in real time.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

9.

At a teaching hospital with a well-known simulation lab, staff rarely call the rapid response team because many nurses believe that calling for help makes them look incompetent. The hospital cafeteria recently extended its hours. At which level of the iceberg model is this belief?

- A. Event.
- B. Structure.
- C. Pattern.
- D. Mental model.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

10. [TRANSFER — reversed stem / different variable]

“Patient falls on our unit have risen every winter for three years.” Which iceberg level does this statement describe?

- A.** Structure.
- B.** Mental model.
- C.** Pattern.
- D.** Event.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

11.

A hospital whose CEO was featured in a trade magazine covers chronic nurse shortages with mandatory overtime every month. Over two years, recruiting efforts stall because overtime “works,” and managers lose their skill at planning staffing. Which archetype does this show?

- A.** Shifting the burden.
- B.** Limits to growth.
- C.** A reinforcing loop of success, since overtime solved the shortage.
- D.** Fixes that fail.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

12.

In a health network with an award-winning website, the emergency department meets its door-to-discharge target, the lab meets its turnaround target, and the inpatient units meet their bed-turnover target. Yet 30-day readmissions have risen. The network added two new board members this year. What systems idea best explains this?

- A.** If every department meets its target, the system must be performing well.
- B.** The readmission measure is wrong because the other metrics are good.
- C.** The new board members disrupted operations.
- D.** Suboptimization: outcomes emerge from how parts interact, not from each part’s score.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

STOP

Answers and dissection follow. Finish every item and every strip first.

Key and dissection

Struck-through text is the camouflage: facts that do not decide the item.

1. Answer A Concept 1

A rural clinic ~~with a new parking lot~~ replaced its entire front-desk team after long wait times, but waits were back within three months. Scheduling still books every patient in 15-minute slots, and pay bonuses still reward visit volume. ~~The clinic's logo was also redesigned.~~ What best explains why the problem returned?

Target word: why new staff did not change behavior **Decides:** The rules and incentives (interconnections, purpose) stayed the same.

A	Correct	Swapping elements rarely changes a system whose connections and goals are the same.
B	Adjacent category	Blames elements; the same result with different people points to structure.
C	Absolute language	Time does not help if the structure keeps producing the problem.
D	Camouflage	The redesign does not affect wait times.

2. Answer D Concept 2

At a hospital ~~that recently repainted its lobby~~, nurse burnout leads to resignations, which raise the workload on remaining nurses, which increases burnout further. ~~The chief nursing officer is new to the region.~~ What kind of loop is this?

Target word: loop type **Decides:** Each step pushes the next in the same direction.

A	Dropped element	Ignores that workload feeds back to burnout, closing the loop.
B	Adjacent category	Delays can exist in any loop; they are not a loop type.
C	Inverted operation	Balancing loops seek a goal; loop type depends on direction of effect, not on good or bad.
D	Correct	Burnout amplifies itself through turnover and workload—a vicious cycle.

3. Answer C Concept 2 · transfer

Which example describes a balancing loop?

Target word: identify a balancing loop — reverse direction **Decides:** The loop pulls the system back toward a target.

A	Confusion set	That amplifies change—reinforcing.
B	Inverted operation	Growth feeding growth is reinforcing.
C	Correct	The response counters the change and moves toward a set point.
D	Absolute language	Delays occur in both loop types.

4. Answer B Concept 3

A hospital ~~with a highly rated cardiac unit~~ starts Monday with 180 inpatients. For the next 4 days it averages 45 admissions and 50 discharges per day. ~~The emergency department also saw 300 visits a day.~~ How many inpatients will there be at the end of day 4?

Target word: stock level after 4 days **Decides:** Net flow = $45 - 50 = -5$ per day.

A	Inverted operation	Adds the net outflow instead of subtracting it.
B	Correct	$180 + (45 - 50) \times 4 = 180 - 20 = 160$.
C	Dropped element	Applies one day of net flow and drops the 4-day time span.
D	Camouflage	Admissions already capture the inflow; ED visits are not the stock's flow.

5. Answer B Concept 3 • transfer

A unit has 180 patients and discharges 50 per day. What average daily admission rate would raise the census to 200 in 4 days?

Target word: solve for inflow **Decides:** Needed net inflow = $(200 - 180) / 4 = 5$ per day.

A	Dropped element	That is the net change; admissions must also replace the 50 discharges.
B	Correct	Outflow 50 + net gain 5 = 55.
C	Inverted operation	Subtracts the net gain from outflow, which would shrink the census.
D	Wrong stem number	Adds the 20-patient total gain to outflow as if it were a daily figure.

6. Answer C Concept 4

A health system ~~known for its research grants~~ sees staffing shortages and posts many jobs. New hires take about 10 weeks to start. Seeing no change after 4 weeks, the manager posts even more jobs; three months later the units are overstaffed and hiring is frozen, and the cycle repeats. ~~The HR office moved to a new building last year.~~ What best explains the swings?

Target word: cause of oscillation **Decides:** A long delay between action and visible result.

A	Absolute language	Delays lead most decision-makers to overcorrect; the fix is structural.
B	Adjacent category	Staffing swings up and down around a target—a delayed balancing loop, not runaway growth.
C	Correct	Acting on old information during a delay produces overshoot and oscillation.
D	Camouflage	The move does not create the hiring delay.

7. Answer A Concept 5

A county health department ~~with a popular newsletter~~ wants to reduce missed prenatal visits. Its partner clinics are paid per visit completed. Which intervention is likely to have the greatest leverage?

Target word: highest-leverage intervention **Decides:** Changing the goal and rules beats adjusting a number.

A	Correct	Changing goals and rules reshapes behavior across the system.
B	Absolute language	Resources are parameters; they rarely outperform changes to goals or rules.
C	Camouflage	Uses the newsletter detail; a small information tweak that leaves incentives unchanged.
D	Adjacent category	A parameter change—the lowest-leverage kind.

8. Answer B Concept 5 • transfer

Which intervention is the lowest-leverage type of change?

Target word: lowest leverage — reverse direction **Decides:** Parameters (numbers) sit at the bottom.

A	Inverted operation	Changing the goal or paradigm is high leverage.
B	Correct	Adjusting a number leaves goals, rules, and mindsets unchanged.
C	Adjacent category	Rule changes rank above parameter changes.
D	Confusion set	New information flows are mid-to-high leverage.

9. Answer D Concept 6

At a teaching hospital ~~with a well-known simulation lab~~, staff rarely call the rapid response team because many nurses believe that calling for help makes them look incompetent. ~~The hospital cafeteria recently extended its hours.~~ At which level of the iceberg model is this belief?

Target word: iceberg level **Decides:** A shared belief that sustains behavior.

A	Inverted operation	Events are the visible tip; this is the deepest level.
B	Adjacent category	Structures are policies, workflows, and physical systems; the belief keeps them in place but is not one.
C	Confusion set	Patterns are trends over time, such as rising late calls.
D	Correct	Beliefs and assumptions sit at the deepest level of the iceberg.

10. Answer C Concept 6 · transfer

“Patient falls on our unit have risen every winter for three years.” Which iceberg level does this statement describe?

Target word: classify a statement — reverse direction **Decides:** A trend over time.

A	Adjacent category	Structure would be the cause, such as winter staffing policy.
B	Confusion set	No belief or assumption is described.
C	Correct	A repeated trend over time is a pattern.
D	Dropped element	Drops the three-year time element; a single fall would be an event.

11. Answer A Concept 7

A hospital ~~whose CEO was featured in a trade magazine~~ covers chronic nurse shortages with mandatory overtime every month. Over two years, recruiting efforts stall because overtime “works,” and managers lose their skill at planning staffing. Which archetype does this show?

Target word: archetype **Decides:** A symptomatic fix erodes capacity for the fundamental solution.

A	Correct	Reliance on overtime weakens the fundamental fix—recruiting and planning.
B	Adjacent category	Describes growth that hits a constraint, not dependence on a quick fix.
C	Inverted operation	Overtime masks the shortage; it does not solve it.
D	Confusion set	That archetype centers on side effects of the fix making the original symptom worse, not on eroding the fundamental solution.

12. Answer D Concept 8

In a health network ~~with an award-winning website~~, the emergency department meets its door-to-discharge target, the lab meets its turnaround target, and the inpatient units meet their bed-turnover target. Yet 30-day readmissions have risen. ~~The network added two new board members this year.~~ What systems idea best explains this?

Target word: why part-level success fails the whole **Decides:** Each part optimizes its own metric at the expense of the whole.

A	Absolute language	The rising readmissions disprove this.
B	Inverted operation	Uses the part metrics to reject the whole-system result.
C	Camouflage	Board changes are unrelated to the metric conflict.
D	Correct	Rushing discharges to hit targets can push problems downstream into readmissions.

Triage grid

Write each item number in the box it belongs to.

	Right	Wrong
Sure	Owned.	Concept inversion — top priority. A wrong mental model; it survives rewording.
Unsure / guess	Recognition, not mastery. Re-drill with the transfer.	Not learned yet.

Trap tally

For every miss, find the letter you chose in the key and tally its trap. The biggest number is the pattern to drill.

Trap	What it does	In this drill	Your misses
Dropped element	Leaves out a required term or step.	4	
Wrong stem number	A figure copied from the prompt that is not the answer.	1	
Adjacent category	The right idea in the wrong bucket.	8	
Inverted operation	Direction flipped: add/subtract, cause/effect, higher/lower.	8	
Absolute language	Always, never, guarantee — where judgment applies.	5	
Confusion set	Every option is real; the test is exact role-matching.	5	
Camouflage	Answers a stem fact that was never the question.	5	

Anchor rows — the concept list

One row per concept: the rule to lock in, the trap that catches people, and the tell in the question that signals it.

#	Concept	Rule	Trap	Tell
1	Elements, interconnections, purpose	A system is elements joined by interconnections to serve a purpose; changing the purpose or the connections changes behavior far more than swapping the elements.	Assuming new people will fix a problem the structure keeps producing.	"We replaced the staff but the problem came back."
2	Reinforcing vs. balancing feedback loops	A reinforcing loop amplifies change in the same direction (growth or collapse); a balancing loop pushes the system toward a goal or limit.	Calling any loop with bad outcomes "balancing," or any loop with growth "good."	A chain of causes that returns to its starting variable.
3	Stocks and flows	A stock changes only through its flows: $\text{change in stock} = (\text{inflow} - \text{outflow}) \times \text{time}$; a stock rises whenever inflow exceeds outflow, even if inflow is falling.	Thinking a stock falls as soon as its inflow drops.	A starting level plus rates per day, week, or month.
4	Delays cause overshoot and oscillation	When feedback arrives late, decision-makers keep correcting and overshoot the goal, producing swings.	Blaming swings on bad people rather than a delay in the loop.	Actions that take weeks to show results, and repeated overcorrection.
5	Leverage points	Changing a system's goals, rules, information flows, or mindset has more leverage than adjusting numbers (parameters) like budgets or quotas.	Assuming the biggest budget change is the biggest lever.	"Which intervention will have the most lasting effect?"
6	The iceberg model	Events are visible; beneath them are patterns over time, the structures that cause the patterns, and the mental models that sustain the structures.	Treating a repeated trend as a single event, or a belief as a structure.	"Which level of the iceberg does this describe?"
7	Fixes that fail vs. shifting the burden	Fixes that fail: a quick fix relieves a symptom but creates side effects that make the problem return worse. Shifting the burden: reliance on a symptomatic fix erodes the capacity for the fundamental solution.	Mixing up the two archetypes, or calling any quick fix a failure.	A short-term solution that becomes a habit or backfires.
8	Emergence and suboptimization	System outcomes emerge from interactions among parts; optimizing each part separately can make the whole worse.	Believing that if every department hits its own target, the system must perform well.	Every unit meets its metric while patient outcomes decline.

Course scope (public description): Introduction to Systems Thinking for Health Professionals provides learners with the skills required to engage in a holistic systems-based approach to analyzing complex problems and solutions. This course introduces the foundational concepts and principles of systems thinking and provides

opportunities to use a systems thinking approach to analyze and evaluate real-world case studies. The course will culminate with using systems thinking to develop a solution to an authentic complex problem. This course has no prerequisites, but general education math (C955 or C957) is preferred. Because the course is self-paced, learners may move through the material as quickly or as slowly as needed, with the goal of demonstrating proficiency in the five competencies covered in the final assessment. If learners have no prior knowledge of this material, they can expect to spend 30 to 40 hours on the course content.